

Triple-Negative Breast Cancer (MONDO:0005494): A Comprehensive Disease Characteristics Report

Summary

Triple-negative breast cancer (TNBC) is an aggressive, molecularly heterogeneous subtype of invasive breast carcinoma that accounts for approximately **10–15% of all new breast cancer diagnoses**. It is operationally defined by the absence of three receptors on tumor immunohistochemistry: estrogen receptor (ER), progesterone receptor (PR), and lack of amplification/overexpression of human epidermal growth factor receptor 2 (HER2). This receptor-negative phenotype removes the two most successful therapeutic targets in breast oncology — endocrine therapy and HER2-directed agents — leaving cytotoxic chemotherapy, and more recently immunotherapy and biology-guided targeted agents, as the therapeutic backbone. TNBC disproportionately affects younger women, women of African ancestry, premenopausal women, and germline *BRCA1* mutation carriers, and it is a major contributor to racial disparities in breast cancer mortality.

Mechanistically, TNBC is best understood as a **TP53-mutant, homologous-recombination-deficient (HRD) disease**. Near-universal *TP53* mutation (with combined loss-of-function and oncogenic gain-of-function) cooperates with loss of homologous-recombination (HR) DNA repair — arising from germline or somatic *BRCA1/2*, *PALB2*, and *RAD51* pathway defects, or from epigenetic *BRCA1* promoter hypermethylation ("BRCAness") — to produce profound genomic instability. This drives a highly proliferative, basal-like tumor cell that undergoes early hematogenous spread with a strong tropism for visceral organs and brain. Superimposed metabolic reprogramming (Warburg glycolysis and glutamine addiction) supports rapid proliferation and stress resistance. TNBC resolves into four tumor-intrinsic transcriptomic subtypes — **basal-like 1 (BL1)**, **basal-like 2 (BL2)**, **mesenchymal (M)**, and **luminal androgen receptor (LAR)** — that differ in biology, prognosis, and therapy response.

Clinically, the HRD/genomic-instability biology has become therapeutically actionable. Standard of care now includes **neoadjuvant pembrolizumab plus chemotherapy (KEYNOTE-522)** for stage II–III disease, **PARP inhibitors (olaparib, talazoparib)** exploiting

synthetic lethality in germline-*BRCA*-mutated tumors, the **Trop-2 antibody–drug conjugate sacituzumab govitecan** for metastatic disease, and **androgen-receptor blockade** for the LAR subtype. Tumor-infiltrating lymphocytes (TILs) predict chemoimmunotherapy response. Despite these advances, metastatic TNBC still carries a dismal prognosis (~12-month median overall survival), and prevention in hereditary disease relies on risk-reducing surgery and intensified surveillance in *BRCA1/2* carriers. This report synthesizes 14 confirmed findings across 65 reviewed papers into the full 15-section disease knowledge template. No primary patient-level data files were provided; all statements are aggregated disease-level evidence from primary literature and reviews.

Key Findings

1. Disease Information

TNBC is a subtype of invasive breast carcinoma "with a poor prognosis and aggressive behavior that accounts for 10-15% of all new cases of breast cancer... characterized by the absence of progesterone and estrogen receptor expression and lacks gene amplification or overexpression of HER2" (PMID: 37044296). It is a molecular/immunohistochemical classification rather than a distinct histologic entity — most TNBCs are invasive ductal carcinomas of no special type, frequently high-grade (grade 3) and basal-like.

Key identifiers and ontology terms:

Resource	Identifier
MONDO	MONDO:0005494 (triple-negative breast cancer)
MeSH	Triple Negative Breast Neoplasms (D064726)
ICD-10	C50.x (malignant neoplasm of breast; TNBC not separately coded)
ICD-O morphology	8500/3 (invasive ductal carcinoma NST, most common)
NCIT	C71732 (Triple-Negative Breast Carcinoma)
Disease Ontology	DOID:0060081

Synonyms / alternative names: triple-negative breast carcinoma; ER-/PR-/HER2- breast cancer; basal-like breast cancer (overlapping but not identical — see below); ductal breast carcinoma triple-negative.

Source of information: This report is derived predominantly from *aggregated disease-level resources* — clinical trials, molecular profiling cohorts (TCGA), registry data (SEER), and mechanistic literature — rather than individual EHR records.

2. Etiology

Causal factors and risk factors. TNBC arises from acquired and inherited defects in genome maintenance. Established epidemiological risk factors are well characterized: "Risk factors for TNBC include young age at breast cancer diagnosis, young age at menarche, high parity, lack of breast feeding, high body mass index and African American ethnicity. The majority of BRCA1 tumours are TNBC" ([PMID: 21080741](#)).

- **Genetic risk factors:** Germline pathogenic variants confer graded risk. In multigene panel testing, "Germline pathogenic variants in BARD1, BRCA1, BRCA2, PALB2, and RAD51D were associated with high risk (odds ratio > 5.0) of TNBC," while BRIP1, RAD51C, and TP53 conferred moderate risk (OR > 2) ([PMID: 30099541](#)). *BRCA1* is the dominant hereditary driver, and the majority of *BRCA1*-associated tumors are triple-negative.
- **Environmental/reproductive risk factors:** young age at menarche, high parity, lack of breastfeeding, high BMI/obesity, and African American ethnicity ([PMID: 21080741](#)).
- **Protective factors:** Breastfeeding is protective (its absence is a risk factor). In *BRCA* carriers post-oophorectomy, estrogen-only hormone replacement therapy was **not** associated with increased breast cancer risk and each year was associated with reduced risk (HR 0.90, 95% CI 0.81–0.99; and HR 0.87 in *BRCA1* carriers) ([PMID: 41949865](#)).
- **Gene–environment interactions:** Obesity and ancestry interact with tumor immunobiology — in a balanced Black/White TNBC cohort, immune deconvolution showed higher abundance of several immune cell populations in tumors from Black women, though neither Black race nor obesity was intrinsically predictive of poor outcome after balancing for stage and subtype ([PMID: 41224793](#)).

3. Phenotypes

TNBC most commonly presents as a **palpable breast mass** (HP:0031416, breast mass), often growing rapidly. In an imaging cohort, TNBC was associated with presentation on breast exam (56%), palpability (59.1%), more invasive ductal carcinoma (92.4%), higher stage (stage III, 16.5%), and higher grade (grade 3, 82.7%), and notably *lower* mammographic breast density (PMID: 31927471).

Phenotype	HPO suggestion	Characteristics
Breast mass / palpable lump	HP:0031416	Adult-onset; often rapidly enlarging; frequently interval (between screens)
High tumor grade (grade 3)	—	82.7% grade 3; severe/aggressive
Visceral metastasis	HP:0030075 (neoplasm of lung), liver/lung involvement	Early; poor prognosis
Brain metastasis	HP:0030692 (brain neoplasm)	Overrepresented in rapid-relapse TNBC (10.9%)
Lymph node metastasis	HP:0002716	Common at presentation

Age of onset: adult, skewing younger than other breast cancers (basal subtype significantly more common in patients aged <50 and premenopausal women; PMID: 37482851). **Severity/progression:** severe, progressive, with early distant relapse. **Quality of life:** In metastatic disease, PARP inhibition improved health-related QoL versus chemotherapy (global health status difference 7.5 points, 95% CI 2.48–12.44, P=0.0035; PMID: 31446213).

4. Genetic / Molecular Information

TP53 and BRCA1/2 are the dominant genomic drivers. In germline *BRCA*-mutated early TNBC (NEOTALA/talazoparib cohort), "Ninety-eight percent of patients had TP53 mutations," and "Strong concordance (97.8%) was observed between tumor *BRCA* and germline *BRCA* mutations, and 90.5% (38/42)... were predicted to exhibit *BRCA* loss of heterozygosity (LOH)" ([PMID: 38869771](#)).

- **Causal genes:** *TP53* (OMIM 191170), *BRCA1* (OMIM 113705), *BRCA2* (OMIM 600185), *PALB2*, *RAD51C/D*, *BARD1*, *BRIP1* (HGNC-approved symbols). High-risk (OR>5): *BARD1*, *BRCA1*, *BRCA2*, *PALB2*, *RAD51D*; moderate-risk (OR>2): *BRIP1*, *RAD51C*, *TP53* ([PMID: 30099541](#)).
- **Somatic vs germline:** *BRCA* alterations occur both germline and somatically, with distinct biological/clinical patterns; somatic alterations "should not be interpreted in isolation and require integration with germline status" ([PMID: 42492369](#)). Whole-exome sequencing of Indian TNBC found ~106 mutations/sample, recurrent somatic mutations in *CTNNB1* (47%), *TP53* (33%), *PIK3CA*, *BRCA1/2*, and mutational signature 3 (HR-repair failure) associated with high tumor mutational burden ([PMID: 41324770](#)).
- **Variant classification/type:** Pathogenic germline *BRCA1* founder-type frameshifts (e.g., c.66_67delAG [185delAG], c.3226_3227delAG) reported in TNBC; somatic *TP53* missense predominate ([PMID: 34097676](#)).
- **Functional consequences:** *BRCA1/2* loss = loss of function → HRD; mutant *TP53* combines loss of function with oncogenic gain of function (see Finding 13 below).
- **Epigenetic information:** "Aberrant methylation patterns are characterized by promoter hypermethylation of tumor suppressor genes (*BRCA1*, *CDH1*, *PTEN*, *RARβ*)... and global hypomethylation that activates oncogenes and promotes genomic instability" ([PMID: 40925466](#)). *BRCA1* promoter hypermethylation phenocopies *BRCA1* loss ("BRCAness").
- **Chromosomal abnormalities:** Genomic instability, chromosomal instability, centrosome amplification, and copy-number alterations mirroring TCGA are characteristic ([PMID: 33267874](#); [PMID: 30859564](#)).

5. Environmental Information

- **Lifestyle factors:** high BMI/obesity, high parity, lack of breastfeeding, and young age at menarche are the established modifiable/reproductive contributors ([PMID: 21080741](#)).
- **Environmental/social determinants:** Residential racialized economic segregation independently affects outcomes — patients in the most deprived counties had higher breast

cancer-specific mortality (HR 1.14) and late-stage diagnosis (OR 1.15), explaining ~28% of the excess Black-vs-White mortality ([PMID: 40105752](#)).

- **Infectious agents:** Not applicable — TNBC is not caused by an infectious agent. (Iatrogenic infection risk, e.g., parvovirus B19 during immunochemotherapy, is a treatment complication, not a cause; [PMID: 42500288](#).)

6. Mechanism / Pathophysiology

Causal chain (initiating lesion → clinical manifestation):

1. Germline or somatic loss of BRCA1/BRCA2/PALB2/RAD51 pathway
–OR– epigenetic BRCA1 promoter hypermethylation ("BRCAness")
| leads to
▼
2. Homologous-recombination deficiency (HRD) – inability to accurately repair DNA double-strand breaks
| results in
▼
3. Near-universal TP53 mutation (loss-of-function + gain-of-function), abrogating G2/M checkpoint and apoptosis
| leads to
▼
4. Genomic instability: chromosomal instability, centrosome amplification, loss of heterozygosity, high mutational burden
| drives
▼
5. Emergence of a highly proliferative, poorly differentiated basal-like tumor cell (EGFR+, CK5/6+, high Ki67)
| supported by
▼
6. Metabolic reprogramming – Warburg aerobic glycolysis + glutamine addiction – fueling biosynthesis, redox/ferroptosis defense
| enables
▼
7. Invasion, EMT, and early hematogenous dissemination
| results in
▼
8. Visceral- and brain-tropic metastasis, chemoresistant relapse, and death (~12-month median OS in metastatic disease)

BRANCH (LAR subtype): androgen-receptor-driven signaling defines a distinct, chemo-resistant, AR-targetable trajectory.

Molecular detail.

- **Mutant TP53 gain-of-function.** "the R181C mutation leads to 'loss-of-function' of transcriptional regulating tumor suppressor genes like p21, bax, and PUMA as well as 'gain-of-function' of transcriptional regulating tumor promoting genes of PIK3CA, SHC1, SRC and PAK4. This dysregulation promotes genomic instability" (PMID: 41484250). Mutant p53 also drives loss of the wild-type allele: "we observed profound stabilization of mutp53 protein, the loss of p21 expression, the abrogation of G2/M checkpoint, chromosomal instability, centrosome amplification" (via Nek2 upregulation; PMID: 33267874). *GO terms*: GO:0006281 (DNA repair), GO:0000724 (double-strand break repair via homologous recombination), GO:0007049 (cell cycle), GO:0006915 (apoptotic process).
- **HRD synthetic lethality.** "Poly(ADP-ribose) polymerase (PARP) inhibitors exploit this deficiency through synthetic lethality... especially in BRCA1 or BRCA2 mutation carriers" (PMID: 29753961). Basal-like/TNBC tumors "share several histo-molecular characteristics with BRCA1-deficient tumors, including genomic instability and reduced BRCA1 expression beyond germline-mutated cases" (PMID: 42394702).
- **Metabolic reprogramming.** TNBC exhibits the Warburg effect and glutamine addiction: "the addiction of TNBC to Gln could facilitate the proliferation and invasiveness of these cancers. Thus, Gln metabolism inhibitors, such as CB-839, could be applied" (PMID: 34083056). Integrated omics in MDA-MB-231 show glutamine drives reductive TCA lipogenesis/sterol synthesis (SREBF1/2) while glucose "licens[es] chromatin engagement, DNA replication, and mitotic progression" (PMID: 42373230). *CHEBI*: glutamine (CHEBI:28300), glucose (CHEBI:17234), lactate (CHEBI:24996).
- **Immune microenvironment.** TILs shape prognosis and therapy response; the immunomodulatory signal reflects infiltrating lymphocytes and varies by subtype (highest in BL1; PMID: 30312311).
- **Cell types (CL):** basal/luminal-progenitor mammary epithelial cell (CL:0002326, luminal epithelial cell of mammary gland; CL:0000646 basal cell), tumor-infiltrating T lymphocyte (CL:0000084), regulatory T cell (CL:0000815).

7. Anatomical Structures Affected

- **Primary organ:** breast (UBERON:0000310), specifically mammary gland epithelium/ductal-lobular units (UBERON:0002114 duct; mammary gland epithelium).
- **Tissue/cell:** epithelial tissue; basal-like tumors express EGFR and cytokeratin 5/6 and arise from a basal/luminal-progenitor lineage.

- **Secondary organ involvement (metastatic tropism):** lung, liver, bone, lymph nodes, and brain. In a real-world sacituzumab govitecan cohort, metastatic sites were lymph nodes (60.9%), bone (43.0%), lung (42.6%), liver (33.2%), brain (26.2%) ([PMID: 42413328](#)). Rapid-relapse TNBC preferentially presents with visceral (74.9%) and brain (10.9%) metastases ([PMID: 42499497](#)).
- **Subcellular (GO CC):** nucleus (GO:0005634, site of TP53/BRCA DNA-repair defects), centrosome (GO:0005813, amplified), mitochondrion (GO:0005739, metabolic rewiring).
- **Lateralization:** unilateral at presentation (as in most breast cancers), bilateral risk elevated in *BRCA* carriers.
- **UBERON:** breast (UBERON:0000310), mammary gland (UBERON:0001911), lung (UBERON:0002048), liver (UBERON:0002107), brain (UBERON:0000955), bone (UBERON:0002481), lymph node (UBERON:0000029).

8. Temporal Development

- **Onset:** adult-onset, skewing younger/premenopausal; basal subtype significantly more common in patients <50 and premenopausal women ([PMID: 37482851](#)).
- **Progression:** rapid. TNBC is graded by AJCC stage; stage III independently predicts worse recurrence-free survival (HR 2.18) and distant metastasis-free survival (HR 2.43) ([PMID: 42292008](#)).
- **Disease course:** aggressive, progressive, with a characteristic early relapse peak. "Rapid relapse" TNBC (distant metastasis/death within 24 months) is associated with platinum resistance and visceral/CNS tropism ([PMID: 42499497](#)).
- **Critical period / window of opportunity:** the neoadjuvant setting — achieving pathologic complete response (pCR) is the key early prognostic milestone and intervention window (KEYNOTE-522).
- **Remission:** treatment-induced (pCR after neoadjuvant chemoimmunotherapy); spontaneous remission is not a feature.

9. Inheritance and Population

- **Epidemiology:** ~10–15% of all breast cancers ([PMID: 37044296](#)); basal-like ~15–20% of cases ([PMID: 42394702](#)).
- **Inheritance:** the hereditary fraction follows **autosomal dominant** transmission of *BRCA1/2*, *PALB2*, *RAD51* pathway variants with incomplete, age-dependent penetrance; the

disease itself is multifactorial/polygenic. Founder *BRCA1* variants (e.g., 185delAG) occur (PMID: 34097676).

- **Sex ratio:** overwhelmingly female (as with breast cancer generally).
- **Population demographics:** Higher incidence and worse outcomes in Black/African American women and younger women. "Black women experience a 36% higher mortality rate" than White women, partly attributable to higher prevalence of aggressive subtypes such as TNBC (PMID: 42320849). SEER analyses show young Black women with TNBC and Asian women <50 carry elevated hazard ratios (PMID: 41896553). In African American women with invasive breast cancer, 64.1% met NCCN genetic-testing criteria; pathogenic mutations concentrated in *BRCA2* (29.4%) and *BRCA1* (15.7%) (PMID: 33083949).

10. Diagnostics

Core diagnosis is immunohistochemistry demonstrating ER-negative, PR-negative, HER2-negative status (with HER2 confirmed by ISH if IHC equivocal). Within TNBC, basal-like tumors are identified by additional markers: "Basal subtype was significantly more common in patients aged <50 years at diagnosis, premenopausal women, patients with positive nodal status, those with grade III tumours, and patients with Ki67 proliferation marker >20% (p<0.05)" — basal markers being EGFR and CK5/6 (PMID: 37482851). An IHC combination of high CK19 + high Ki-67 + E-cadherin loss identified the worst relapse-free and overall survival (PMID: 39435793).

- **Imaging:** mammography (TNBC associated with lower mammographic density), MRI, ultrasound; TNBC often presents as an interval/palpable mass (PMID: 31927471). Trop-2-directed PET/CT is emerging (PMID: 42559486).
- **Biopsy/pathology:** core-needle biopsy with histopathology (usually invasive ductal carcinoma NST, grade 3) plus IHC panel.
- **Genetic testing:** Germline multigene panel testing (*BRCA1/2*, *PALB2*, *RAD51C/D*, *BARD1*, *BRIP1*, *TP53*) is recommended given the high hereditary yield (PMID: 30099541); tumor *BRCA* and germline *BRCA* show 97.8% concordance (PMID: 38869771). NCCN criteria capture the majority but miss ~12.5% of African American mutation carriers (PMID: 33083949).
- **Omics/transcriptomic diagnostics:** Lehmann TNBCtype-4 subtyping (BL1, BL2, M, LAR) and PAM50, plus HRD/genomic scar scores and DNA-damage-repair signatures, refine classification and predict therapy (PMID: 27310713; PMID: 37334114).
- **Predictive biomarkers:** PD-L1 (immunotherapy), germline/somatic *BRCA* and HRD (PARP inhibitors), TILs (pCR), Trop-2 (ADC), AR (LAR subtype).

11. Outcome / Prognosis

TNBC has the worst stage-adjusted prognosis of the major breast cancer subtypes. **Metastatic TNBC "bears a dismal prognosis with an average survival of 12 months"** (PMID: 29753961). In a large real-world PD-L1-negative metastatic cohort (n=929), "Median rwOS and rwPFS... were 12.2 (10.7, 14.4) months and 4.7 (4.5, 5.2) months" (PMID: 42554864).

Prognostic factor	Direction	Source
Stage III	Worse RFS (HR 2.18), DMFS (HR 2.43)	PMID: 42292008
TILs present	Better (higher pCR: 70% vs 48%)	PMID: 42360368
pCR achieved	Better EFS/OS	PMID: 42607489
Basal markers (EGFR/CK5/6, CK19+Ki67+/E-cad loss)	Worse RFS/OS	PMID: 39435793
LAR subtype	Chemo-resistant, lowest pCR	PMID: 29940524
Low SPDEF expression	Worse RFS/OS	PMID: 42305508
High miR-1911-3p	Worse recurrence-free survival (HR 3.04)	PMID: 42517063

Racial disparity: Black women experience 36% higher breast cancer mortality, and residential segregation explains ~28% of the excess mortality (PMID: 42320849; PMID: 40105752). **Complications** include visceral organ failure from metastasis, CNS metastasis, and treatment toxicities (e.g., immune-related adverse events, cytopenias).

12. Treatment

TNBC treatment has shifted from chemotherapy-only to **biology-guided, multimodal therapy**.

Neoadjuvant chemoimmunotherapy (standard for stage II–III). "Neoadjuvant chemotherapy (NACT) combined with pembrolizumab is now standard care for early-stage, triple-negative breast cancer (TNBC), based on KEYNOTE-522 demonstrating improved

pathological complete response (pCR), event-free survival, and overall survival" (PMID: 42607489). Real-world pCR ~57%, and TILs predict response (70% vs 48% pCR, $p=0.0027$; TIL-positive 2.44× more likely to achieve pCR; PMID: 42360368). NCIT: pembrolizumab (C106432).

PARP inhibitors (germline BRCA). In OlympiAD, olaparib prolonged PFS versus chemotherapy, with strong benefit in the triple-negative subgroup (PFS HR 0.47, 95% CI 0.32–0.69; PMID: 36971103), durable long-term survival in first-line metastatic disease (PMID: 36893711), and improved QoL (PMID: 31446213). Talazoparib is used in the neoadjuvant germline-BRCA setting (PMID: 38869771). NCIT: olaparib (C71721), talazoparib (C97316).

Trop-2 antibody–drug conjugate (metastatic). Sacituzumab govitecan (SG) delivers durable benefit in real-world mTNBC: median rwPFS 6.4 months, rwOS 13.6 months, ORR/CR+PR 37.5% (PMID: 42413328). Combination with PD-1 inhibitors ± antiangiogenics improves response in later lines (PMID: 42714605), and SG synergizes with TRAIL agonists preclinically (PMID: 42601637). NCIT: sacituzumab govitecan (C124255).

Androgen-receptor blockade (LAR subtype). "The partial AR antagonist bicalutamide and the next-generation AR inhibitor enzalutamide are being assessed in standard protocols for the mitigation of TNBC" (PMID: 36612226). AR blockade induces G1 arrest in AR-positive MDA-MB-453 cells; nuclear AR is detectable in ~30% of TNBC (PMID: 42373057). The LAR subtype is chemo-resistant with the lowest pCR (PMID: 29940524). NCIT: bicalutamide (C1685), enzalutamide (C95733).

Other modalities: cytotoxic chemotherapy backbone (anthracycline/taxane, platinum for HRD), surgery (breast-conserving or mastectomy), and adjuvant radiotherapy. Experimental: metabolic inhibitors (CB-839/telaglenastat targeting glutaminase; PMID: 34083056), p53-reactivating small molecules (PRIMA-1/APR-246; PMID: 35234267), PLK2 inhibitors to disarm mutant p53 (PMID: 39480521), and iPSC-derived mesothelin-targeted CAR-NK cells (PMID: 37584622).

Adverse events: SG toxicities include alopecia (69.1%), neutropenia (57%), nausea, anemia, diarrhea (PMID: 42413328); pembrolizumab carries immune-related adverse events and rare severe cytopenias (PMID: 42500288; PMID: 42314326).

13. Prevention

- **Primary prevention (hereditary):** For *BRCA1/2* carriers, guideline strategies include risk-reducing mastectomy, risk-reducing bilateral salpingo-oophorectomy (RRBSO), and intensified MRI surveillance. In a population-based cohort, "Of those without a prior breast

cancer (n = 1021), 20.8% underwent bilateral mastectomy... 71.8% underwent RRBSO" (PMID: 42166900); breast MRI uptake was ~31–33% (PMID: 42647762).

- **Secondary prevention:** breast screening (mammography/MRI in high-risk women); TNBC's lower mammographic density and interval-cancer tendency underscore the value of supplemental MRI in carriers.
- **Genetic counseling and cascade testing:** identify carriers for prevention; decision-coaching reduces decisional conflict and increases decision-making in carriers with open family planning (PMID: 42607047). Disparities in counseling access persist for young Black women (PMID: 25868867).
- **Behavioral/public-health:** weight management, breastfeeding, and addressing socioeconomic/segregation drivers of late-stage diagnosis (PMID: 40105752).

14. Other Species / Natural Disease

Spontaneous triple-negative mammary carcinoma occurs naturally in companion animals, providing immunocompetent comparative models. "Feline invasive mammary carcinomas are characterized by their high clinical aggressiveness, rare expression of hormone receptors, and pathological resemblance to human breast cancer, especially triple-negative breast cancer" — including an immune-suppressed, FOXP3⁺ Treg-infiltrated basal-like subgroup mirroring human TNBC immunobiology (PMID: 31959092). In dogs, mammary epitheliosis "was a slow-growing, triple-negative process with a strong predominance of basal-like nonmyoepithelial cells" (CK14⁺) associated with malignancy (PMID: 35451346).

- **Taxonomy:** *Felis catus* (NCBI:txid9685), *Canis lupus familiaris* (NCBI:txid9615), *Mus musculus* (NCBI:txid10090).
- **Orthologous genes:** *Trp53*, *Brca1*, *Brca2* (mouse orthologs of human *TP53*, *BRCA1*, *BRCA2*).
- **Veterinary relevance:** naturally occurring, immunocompetent, spontaneously arising — valuable for microenvironment and immunotherapy studies.

15. Model Organisms

TNBC is studied across in vitro, in vivo, and ex vivo systems.

Model	Examples	Use
Human cell lines	MDA-MB-231 (basal/mesenchymal), MDA-MB-453 (LAR), MDA-MB-468, HCC1806, HCC1937	Mechanistic, drug testing
Syngeneic mouse	4T1	Immunocompetent metastasis/immunotherapy (PMID: 33727591)
Patient-derived xenografts	61-PDX panel	Recapitulate molecular heterogeneity (PMID: 30859564)
Organoids	Patient-derived	Precision phototherapy/drug testing (PMID: 38532439)
iPSC-derived	CAR-NK (mesothelin-targeted)	Immunotherapy development (PMID: 37584622)

PDX fidelity: "TNBC PDX represent all of the various TNBC subtypes identified by the Lehmann classification except for immunomodulatory subtype, which is underrepresented in PDX. NGS and copy number data showed a similar diversity of significantly mutated gene and somatic copy number alteration in PDX and the Cancer Genome Atlas TNBC patients" ([PMID: 30859564](#)).

Limitation: the immunomodulatory/TIL-driven biology is underrepresented in immunodeficient xenografts — a key gap given the therapeutic importance of TILs and immunotherapy.

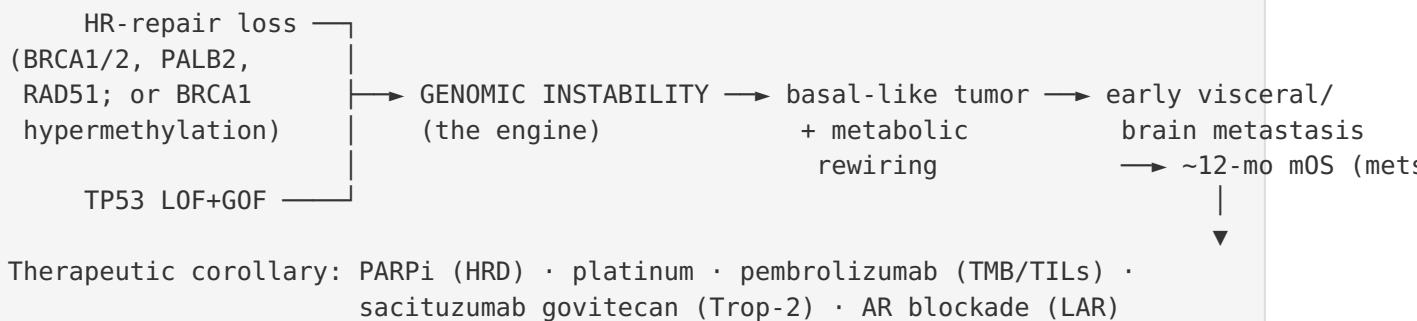
Mechanistic Model / Interpretation

TNBC is best conceptualized not as a single disease but as a **convergent phenotype of genome-maintenance failure** landing on the mammary epithelium. Two upstream lesions dominate and cooperate:

1. **HR-repair collapse** (BRCA1/2/PALB2/RAD51 mutation or *BRCA1* promoter hypermethylation) removes the cell's ability to faithfully repair double-strand breaks.
2. **TP53 loss/gain-of-function** removes the checkpoint that would otherwise arrest or eliminate genomically damaged cells, and actively upregulates oncogenic programs (PIK3CA, SRC, PAK4).

Together these produce runaway **genomic instability** — the disease's central engine — which manifests as chromosomal instability, LOH, centrosome amplification, and high mutational burden. Downstream, this instability generates a basal-like, highly proliferative tumor (EGFR+, CK5/6+, Ki67-high) that is metabolically rewired (Warburg glycolysis + glutamine addiction) to sustain proliferation and resist oxidative/ferroptotic stress. The tumor disseminates early, with a distinctive visceral and brain tropism, producing the disease's poor prognosis.

Crucially, the same instability that makes TNBC aggressive is its **therapeutic Achilles heel**: HRD confers synthetic-lethal sensitivity to PARP inhibitors and platinum, high mutational/neoantigen burden invites checkpoint immunotherapy (potentiated where TILs are present), and surface Trop-2 offers an ADC target. The four transcriptomic subtypes (BL1/BL2/M/LAR) overlay biological trajectories onto this core, with LAR representing a distinct AR-driven, chemo-resistant but AR-targetable branch. The persistent racial disparity is a product of both biology (higher TNBC prevalence in Black and young women) and structural factors (segregation, delayed diagnosis, unequal access to genetic services).



Evidence Base

PMID	Contribution	Supports / Challenges
37044296	Definition, 10–15% prevalence, receptor-negativity	Supports Finding 1
21080741	Epidemiological risk factors, BRCA1 association	Supports Findings 1–2
27310713	TNBCtype-4 subtypes (BL1/BL2/M/LAR)	Supports Finding 2
38869771	98% TP53, 97.8% BRCA concordance, 90.5% LOH	Supports Finding 3
30099541	Panel-based risk genes (OR>5 high-risk)	Supports Finding 3
42607489	KEYNOTE-522 standard of care	Supports Finding 4
42360368	TILs predict pCR (70% vs 48%)	Supports Finding 4
36971103	Olaparib PFS HR 0.47 in TNBC	Supports Finding 5
29753961	PARP synthetic lethality; 12-mo mOS	Supports Findings 5–6
42320849	36% higher Black mortality	Supports Finding 6
42499497	Visceral/brain tropism of rapid relapse	Supports Finding 6
37482851	Basal markers ↔ aggressive features	Supports Finding 7
39435793	CK19+Ki67+/E-cad loss = worst survival	Supports Finding 7
42166900	Risk-reducing surgery uptake	Supports Finding 8
31959092	Feline TNBC model	Supports Finding 9
35451346	Canine basal-like mammary dysplasia	Supports Finding 9
36612226	Bicalutamide/enzalutamide for LAR	Supports Finding 10
29940524	LAR clinical behavior	Supports Finding 10
30859564	PDX fidelity/heterogeneity	Supports Finding 11
40925466	BRCA1 hypermethylation/BRCAness	Supports Finding 12
41484250	Mutant p53 LOF+GOF	Supports Finding 13
33267874	p53LOH, CIN, centrosome amplification	Supports Finding 13
34083056	Glutamine addiction, CB-839	Supports Finding 14
42373230	Glucose/glutamine metabolic states	Supports Finding 14

Limitations and Knowledge Gaps

- **No primary dataset was analyzed:** This report is a literature/knowledge synthesis; no de novo statistical analysis of a patient cohort or omics dataset was performed. Effect sizes and quotes are drawn from published cohorts of varying size, geography, and quality (several are single-center or real-world retrospective studies).
- **Subtype instability:** The Lehmann TNBCtype-4 classification is refined but still evolving; immunomodulatory (IM) and mesenchymal stem-like (MSL) signals were reassigned to microenvironment rather than tumor-intrinsic biology, and boundaries between "basal-like" (PAM50) and "TNBC" (IHC) are imperfect (~partial overlap).
- **Model gaps:** Immunodeficient PDX/xenograft models underrepresent the immunomodulatory/TIL-driven biology that is central to modern immunotherapy — a major translational blind spot.
- **HRD quantification:** The exact prevalence and best clinical assay for HRD beyond germline *BRCA* (e.g., genomic scar scores, *BRCA1* methylation) remain incompletely standardized.
- **Disparities causation:** The relative contributions of ancestry-linked tumor biology versus structural/socioeconomic factors to racial outcome disparities are entangled and not fully resolved.
- **Metastatic prognosis:** Despite new agents, metastatic TNBC survival remains ~12 months; predictive biomarkers for immunotherapy (beyond PD-L1/TILs) and ADC resistance mechanisms are incompletely defined.

Proposed Follow-up Experiments / Actions

1. **Quantitative HRD landscape:** Analyze a genomic dataset (e.g., TCGA-BRCA TNBC subset or a public cohort) to quantify the fraction of TNBC that is HRD-positive by combined germline *BRCA*, somatic *BRCA/PALB2/RAD51*, and *BRCA1* methylation, and correlate with genomic scar scores — filling the standardization gap.
2. **Subtype × therapy-response mapping:** Integrate TNBCtype-4 subtype calls with pCR and survival across neoadjuvant cohorts to formally test whether BL1 (immune-rich) predicts pembrolizumab benefit and LAR predicts AR-blockade benefit.

3. **Immune-competent modeling:** Prioritize syngeneic (4T1), humanized-mouse, or organoid-immune co-culture systems to study checkpoint/ADC combinations, addressing the PDX immunobiology gap.
 4. **Metabolic-DNA-repair combination trials:** Preclinically test glutaminase inhibition (CB-839) or p53-reactivation (APR-246/PLK2 inhibition) combined with PARP inhibitors/platinum, exploiting the convergence of metabolic and genomic vulnerabilities.
 5. **Equity intervention study:** Design a prospective analysis linking cascade genetic-testing access, MRI surveillance uptake, and stage-at-diagnosis in Black and young women to quantify how much of the mortality gap is modifiable through service delivery.
 6. **Liquid-biopsy relapse detection:** Evaluate ctDNA (TP53/BRCA-informed) for early detection of rapid-relapse TNBC within the 24-month high-risk window.
-

Report compiled from 14 confirmed findings across 65 reviewed papers over 5 iterations. Evidence source types span human clinical trials/cohorts, model organism (mouse, cat, dog), in vitro cell-line, and computational/omics studies, as annotated throughout.

Generated by OpenScientist — Scientific Hypothesis Agent for Novel Discovery